

Fabio Vargas

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Citizenship: USA Citizen | Brazil Citizen

PROFESSIONAL SUMMARY

AI/ML scientist and physicist with extensive experience developing machine learning, statistical, and deep learning systems for large-scale data environments. Expertise includes time series forecasting, probabilistic graphical models, Bayesian inference, graph machine learning, and generative AI systems using large language models. Strong background translating complex scientific and business problems into scalable data-driven solutions.

Experience spans enterprise AI development, large-scale vehicle telemetry analytics, and scientific research in remote sensing and atmospheric science. Proven track record leading cross-functional teams, building distributed machine learning systems, and deploying production models using cloud platforms and MLOps workflows.

Author of 10+ peer-reviewed scientific publications and inventor on patented machine learning systems.

TECHNICAL EXPERTISE HIGHLIGHTS***Machine Learning Systems***

Design and development of large-scale machine learning systems for enterprise analytics, including forecasting models, graph machine learning architectures, and probabilistic models for decision support.

Time Series Forecasting & Statistical Modeling

Development of forecasting systems using statistical and machine learning approaches including SARIMAX, exponential smoothing models, Bayesian inference methods, and deep learning architectures such as LSTM networks.

Graph Machine Learning & Network Modeling

Design of graph-based machine learning models to represent complex system relationships, including GraphSAGE architectures applied to vehicle infotainment ecosystems and behavioral modeling.

Causal Modeling & Probabilistic Graphical Models

Application of Bayesian networks, Directed Acyclic Graphs, and probabilistic inference techniques to model causal relationships and risk prediction in complex systems.

Generative AI Systems & LLM Applications

Development of enterprise generative AI platforms using large language models, Retrieval Augmented Generation (RAG), LangChain, and LangGraph to support innovation frameworks and decision workflows.

Distributed Machine Learning & Data Platforms

Implementation of large-scale machine learning pipelines using Apache Spark and Azure Databricks for distributed analytics across large telemetry datasets.

Scientific Computing & Signal Processing

Extensive background in scientific data analysis, remote sensing, and signal processing methods including spectral analysis, FFT methods, and atmospheric image analysis.

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL, C++, Fortran, Matlab, LabView, IDL, VBS

- **Machine Learning & AI:** Deep Learning, Graph Machine Learning, Bayesian Inference, Probabilistic Graphical Models, Generative AI, Large Language Models (LLMs), Retrieval Augmented Generation (RAG), Agentic AI Systems
- **Frameworks & Libraries:** PyTorch, TensorFlow, Keras, Scikit-learn, PyMC3, PGMpy, LangChain, LangGraph, NumPy, Pandas, GeoPandas, Matplotlib
- **Distributed Computing & Data Platforms:** Apache Spark, SparkML, Graph ML, Azure Databricks
- **MLOps & Infrastructure:** MLflow, Docker, Airflow, CI/CD pipelines, model deployment and monitoring
- **Analytics & Modeling:** Time Series Forecasting, Causal Inference, A/B Testing, Model Validation, Statistical Modeling
- **Algorithms & Methods:** CNNs, RNNs, GraphSAGE, SVM, XGBoost, Random Forest, Decision Trees, PCA, SVD, Clustering (KMeans, kNN, HDBSCAN), Recommender Systems
- **Visualization & Analytics Tools:** PowerBI

PROFESSIONAL EXPERIENCE

General Motors Company

Senior AI / Machine Learning Scientist | 2021 – Present

Lead development of machine learning and AI systems for large-scale vehicle telemetry analytics, enterprise forecasting systems, and generative AI innovation platforms.

Key contributions include:

- Led development of generative AI agents implementing TRIZ and Six Thinking Hats innovation frameworks using LangChain and LangGraph with Retrieval Augmented Generation.
- Developed deep learning CNN models for automated road construction detection from vehicle telemetry data.
- Designed LSTM and Bi-LSTM architectures for forecasting dealership pricing behavior across multiple U.S. market regions.
- Built graph machine learning models using GraphSAGE to model relationships between vehicle infotainment applications and user behavior.
- Led development of causal inference models using Bayesian networks and Directed Acyclic Graphs to predict roadway safety incidents across the U.S. road network.
- Developed statistical models estimating Annual Average Daily Traffic (AADT) using probe vehicle telemetry datasets.
- Led forecasting initiatives for GM mobile applications using SARIMAX, Exponential Smoothing, Theta, and Prophet models.
- Developed internal analytics libraries incorporating uncertainty and risk analysis into enterprise predictive models.
- Mentored junior data scientists and collaborated across engineering and product teams to ensure solution adoption.
- Implemented machine learning pipelines using Azure Databricks and MLflow model registry.

University of Illinois at Urbana-Champaign

Research Scientist | 2012 – 2021

Conducted research in atmospheric physics, remote sensing, and scientific data analysis.

Key contributions include:

- Applied deep learning methods for atmospheric image classification.

- Developed signal processing pipelines using FFT, cross-spectral analysis, and statistical modeling for atmospheric datasets.
- Designed and maintained LIDAR remote sensing instrumentation and optical systems.
- Developed LIDAR data acquisition software using LabView.
- Published more than 10 peer-reviewed scientific papers in atmospheric and space science journals.

International Facility Manager — NSF Atmospheric Research Program

University of Illinois at Urbana-Champaign

Led operations of an international atmospheric research facility in Chile for 10 years, coordinating collaboration between research institutions across North and South America.

Responsibilities included:

- Managing international scientific collaboration and research operations
- Coordinating multi-institution research programs
- Supporting NSF-funded atmospheric and remote sensing projects
- Managing technical infrastructure and atmospheric observation instrumentation

University of Illinois at Urbana-Champaign

Postdoctoral Research Associate | 2010 – 2012

- Conducted research in atmospheric remote sensing and instrumentation development
- Managed atmospheric measurement systems at a Chilean research facility
- Contributed to multiple peer-reviewed publications

Altran Technologies — Brazil

R&D Consultant | 2008 – 2009

- Provided research and development consulting to engineering teams
- Improved engineering efficiency and product performance through technical optimization initiatives

LEAD AI SCIENTIST — MAJOR MACHINE LEARNING PROJECTS

Served as lead scientist and technical owner for the following machine learning systems developed at General Motors. All projects used **Apache Spark distributed computing**, except where noted.

Image Quality Classifier

- Classify scientific night-sky images into clean and contaminated categories.
- Tools: convolutional neural networks. Note: this project did not use Spark.

Speed Bump Discovery

- Detect geospatial locations of speed bumps using high-speed vehicle telemetry data.
- Tools: support vector machines, spline interpolation, Apache Spark.

Canonical Road Network (CRN)

- Reconstruct road network topology using vehicle telemetry data.
- Tools: HDBSCAN clustering, Apache Spark.

Infotainment Head Unit Graph Modeling

- Graph representation of vehicle infotainment applications, user behavior, and vehicle segments.
- Tools: GraphSAGE, GPUs, Apache Spark.

Road Construction Detection

- Detect construction zones from vehicle telemetry data to support autonomous driving systems.
- Tools: CNN models, GPUs, Apache Spark.

Dynamic Invoice Price Recommender (DIPR)

- Forecast vehicle transaction prices to guide pricing strategies across U.S. market regions.
- Tools: recurrent neural networks, statistical forecasting models, Apache Spark.

Predictive Roadway Incident Safety Model (PRISM)

- Identify accident hotspots across the U.S. road network.
- Tools: Bayesian networks, probabilistic graphical models, generalized linear models, Spark on Databricks.

Platform for Innovative Exploration (PIX)

- Enterprise generative AI platform supporting innovation frameworks.
- Tools: LangChain, LangGraph, LLMs, RAG, Spark on Databricks, MLOps pipelines.

Mobile App Usage Forecaster

- Forecast usage patterns of GM OnStar mobile application features.
- Tools: statistical forecasting models, ensemble methods, Apache Spark on Databricks, MLOps pipelines.

Residual Value Intelligence (RVI)

- Forecast depreciation and auction values of rental vehicles.
- Tools: generalized linear models, Apache Spark on Databricks, MLOps pipelines.

Short Term Sales Forecast (STSF)

- Forecast vehicle market share and production demand.
- Tools: statistical forecasting models, Apache Spark on Databricks, MLOps pipelines.

TECHNICAL LEADERSHIP & SYSTEMS IMPACT

Led development of machine learning systems and analytics platforms supporting large-scale enterprise decision making. Representative contributions include:

- Designing distributed machine learning pipelines using Apache Spark and Databricks
- Deploying machine learning models using MLflow model registry and MLOps workflows
- Integrating predictive models into enterprise decision systems
- Mentoring data scientists and guiding adoption of advanced analytics methods
- Developing internal analytics libraries and reusable modeling frameworks

EDUCATION

University of Illinois at Urbana-Champaign

- Postdoctoral Research — Remote Sensing and Space Sciences | 2010 – 2012

National Institute for Space Research — Brazil

- Ph.D., Space Geophysics | 2003 – 2007
M.Sc., Space Geophysics | 2001 – 2003

Federal University of Santa Maria — Brazil

- B.S., Physics | 1997 – 2000

CERTIFICATIONS

- Deep Learning Specialization — DeepLearning.AI
- Machine Learning — Stanford University
- Introduction to Big Data — UC San Diego
- SQL for Data Science — UC Davis

- Web Application Development with Python and Flask — Coursera

PUBLICATIONS

Author or co-author of multiple peer-reviewed scientific publications in atmospheric science, remote sensing, and space physics.

Full publication list available upon request.

RESEARCH IMPACT, PATENTS & AWARDS

Scientific Recognition

- NASA Henry J. E. Reid Award — Best Paper (2013)
- Top Cited Article — Earth and Space Science Journal (2021–2022)

Patents

- Predictive Roadway Incident Safety Model (PRISM) — Patent, General Motors (2025)

Corporate Trade Secrets

- In-Stream Systems — General Motors Trade Secret (2023)
- Platform for Innovative Exploration (PIX) — General Motors Trade Secret (2025)

PROFESSIONAL AFFILIATIONS

Member — American Geophysical Union (AGU)

LANGUAGES

English	—	Fluent
Portuguese	—	Native
Spanish — Fluent		

REFERENCES

Available upon request